SUPERNOVA × SPACEPOINT · THEME 03 · SEAS OF DATA

Radar sees every ship. We find the ones that hide.



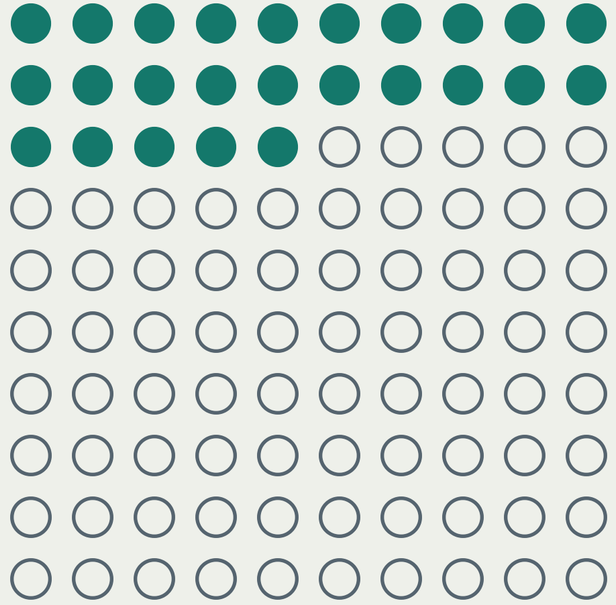
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BITS Pilani, Dubai Campus · November 2026

About three in four industrial fishing vessels are missing from public ship tracking

75%

of industrial fishing vessels are not on public AIS tracking. AIS is a signal each ship chooses to send.

Global figure, 2017 to 2021, not specific to the UAE. Dark does not mean illegal, and radar shows that a ship is there, not who it is.



Each circle = 1 of 100 vessels.
Filled = tracked.

SOURCE: GLOBAL FISHING WATCH, NATURE 2024

Sentinel-1 passes over Jebel Ali about every three days

COVERAGE

250 km swath, 10 m pixels, VV band

CONDITIONS

Day, night and through cloud

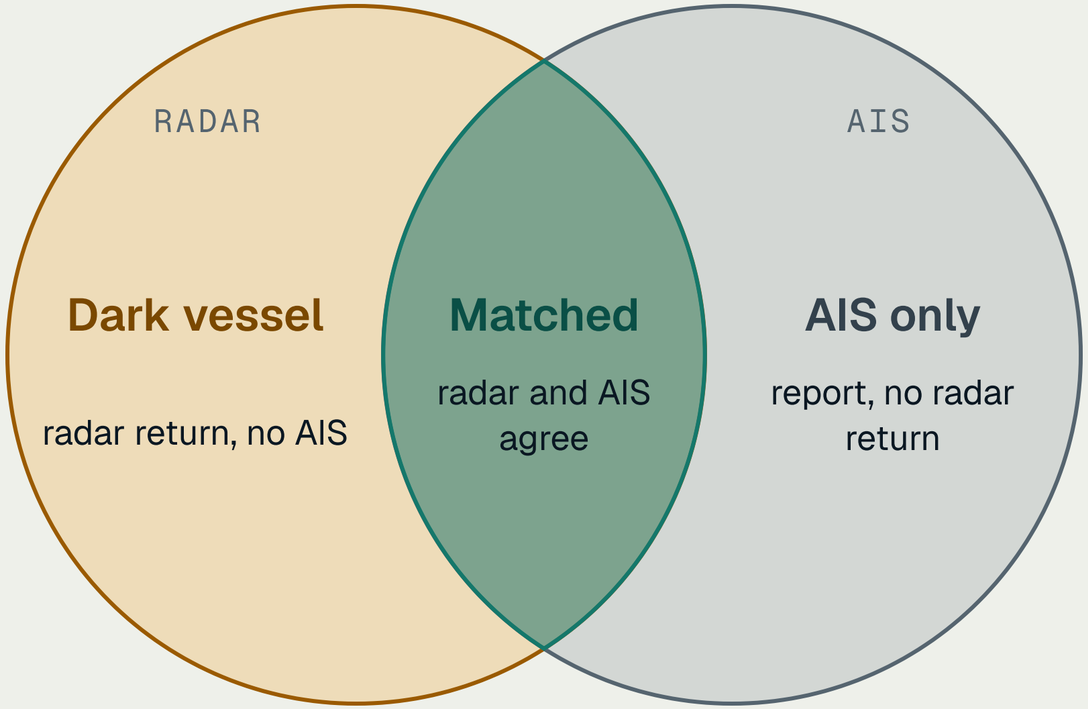
ACCESS

Free and open through Copernicus



PASSES FOUND IN THE COPERNICUS CATALOGUE · 25.01°N 55.06°E

Radar minus AIS equals the dark vessels



THE RULE

A ship is matched when an AIS report lies within 500 m of a radar detection. No report in range: it is flagged dark.

WHY IT HELPS

Dark is a lead, not a verdict. It tells patrols and port authorities where to look first.

IN THIS PROTOTYPE

AIS reports are simulated, because no free real Gulf feed exists. Every output is labelled that way.

What is simulated, what is assumed, and what comes next

Item	Status	Effect on the result
AIS reports	Simulated	Dark share is illustrative until a real feed is connected
Port capacity	Assumed	Scales berth occupancy only; the activity index is unaffected
Training data	Public HRSID	Finer pixels than Sentinel-1; tested on real scenes against a classical detector
Buoy	Design only	CAD model and renders, nothing built

NEXT

1. Connect a real AIS feed
2. Retrain on xView3-SAR, which labels dark vessels
3. Add oil-spill detection on the same radar scenes

Data, models and licences

Component	Used for	Source and terms
Sentinel-1 GRD	Real Gulf scenes, VV band	ESA, via Copernicus Data Space, free and open
HRSID	Training and held-out test of the ship detector	Wei et al., IEEE Access; repository under GPL-3.0
YOLO11	Detector, fine-tuned from pretrained weights	Ultralytics
Port boxes	Counting ships inside Jebel Ali and Khalifa Port	Drawn by us, checked against OpenStreetMap extents

Contains modified Copernicus Sentinel data 2026.